

Comparison of EKF and UKF State Estimation with Measurement Gating and RK4 Propagation

J.C. Vaught

Department of Mechanical Engineering

University of South Carolina

Columbia, South Carolina, USA

jvaught@sc.edu

Abstract—This report addresses the six required tasks in the EMCH792 final project for the nonlinear kinematic bicycle model. The calibration pass yielded $R_1 = 0.087247$ and $R_2 = 0.006536$ with a constant sample period of 0.100 s. The vanilla rectangular comparison favored the UKF over the EKF, reducing position RMSE from 5.1335 m to 2.8483 m and heading RMSE from 22.1313 deg to 4.6895 deg. A fixed scalar 99% normalized-innovation-squared gate with threshold 6.6349 improved heading RMSE by 16.0203 deg for the EKF and 0.9017 deg for the UKF, but position RMSE increased because repeated y_1 rejections made the gated runs more prediction dominated. Replacing rectangular UKF propagation with RK4 reduced position RMSE by 0.2026 m, while the RK4 runtime increased by only 0.960 ms relative to the rectangular UKF. Four appendix ablations test sensitivity to gate threshold, measurement subset, initial covariance scale, and UKF sigma-point spread.

Index Terms—Extended Kalman filter, Unscented Kalman filter, Nonlinear state estimation, Measurement gating, Runge-Kutta integration

I. PROBLEM I. CALIBRATION VARIANCE ESTIMATION

The first task was to estimate the two sensor variances from the calibration data and confirm the deployment sample period. The calibration pass produced $R_1 = 0.087247$ for the range-like measurement and $R_2 = 0.006536$ for the cubic heading sensor. The deployment dataset remained uniformly sampled at 0.100 s. The filters were initialized from a zero-centered prior and used the nominal base covariance so that the baseline comparison did not consume the provided truth channels as estimator inputs.

II. PROBLEM II. VANILLA EKF AND UKF WITH RECTANGULAR INTEGRATION

Both filters used the assignment state $x = [x, y, \psi, u]^T$ with steering angle and longitudinal acceleration as inputs, fixed geometry $l_f = l_r = 1$, and the prescribed continuous-time process covariance. The EKF used rectangular Euler prediction with first-order Jacobian linearization for both the process and measurement updates. The UKF used the scaled unscented transform with $\alpha = 10^{-3}$, $\beta = 2$, and $\kappa = 0$, and propagated every sigma point with the selected discrete-time integrator. In every configuration, the two measurements were processed sequentially as scalar updates so that the gating study in Problem IV could test each sensor independently.

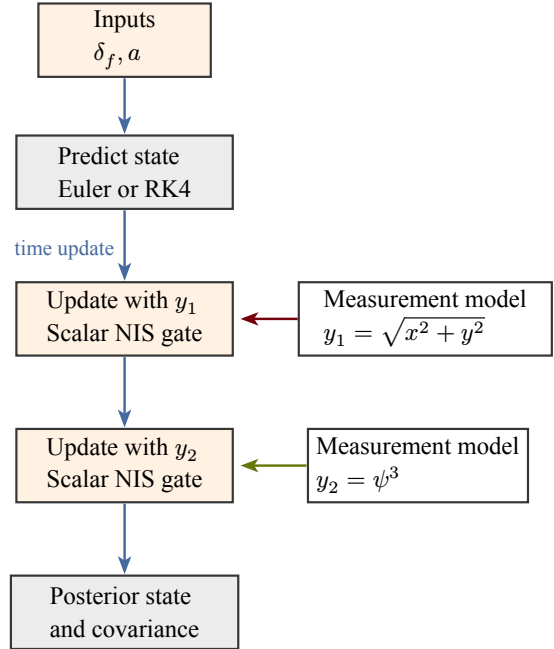


Fig. 1. Standalone CeTZ diagram of the sequential estimation workflow. The figure was authored separately, compiled to PDF, and then imported into the main IEEE-formatted document as a standard image asset.

The ungated rectangular comparison favored the UKF. The EKF rectangular run produced a position RMSE of 5.1335 m with a heading RMSE of 22.1313 deg, while the rectangular UKF reduced these values to 2.8483 m and 4.6895 deg. The same trend appeared in the covariance histories. The rectangular EKF carried an average covariance trace of 0.9432, while the rectangular UKF carried 9.8452. The UKF therefore remained less overconfident on this nonlinear problem, especially under the cubic heading measurement. Appendix B isolates how much of this difference comes from the two measurement channels, and Appendix C checks whether the conclusion is an artifact of the selected initial covariance.

III. PROBLEM III. RUNTIME OF THE VANILLA EKF AND UKF

The average runtime study for the vanilla rectangular filters used sixty repeated executions after one warm-up pass per

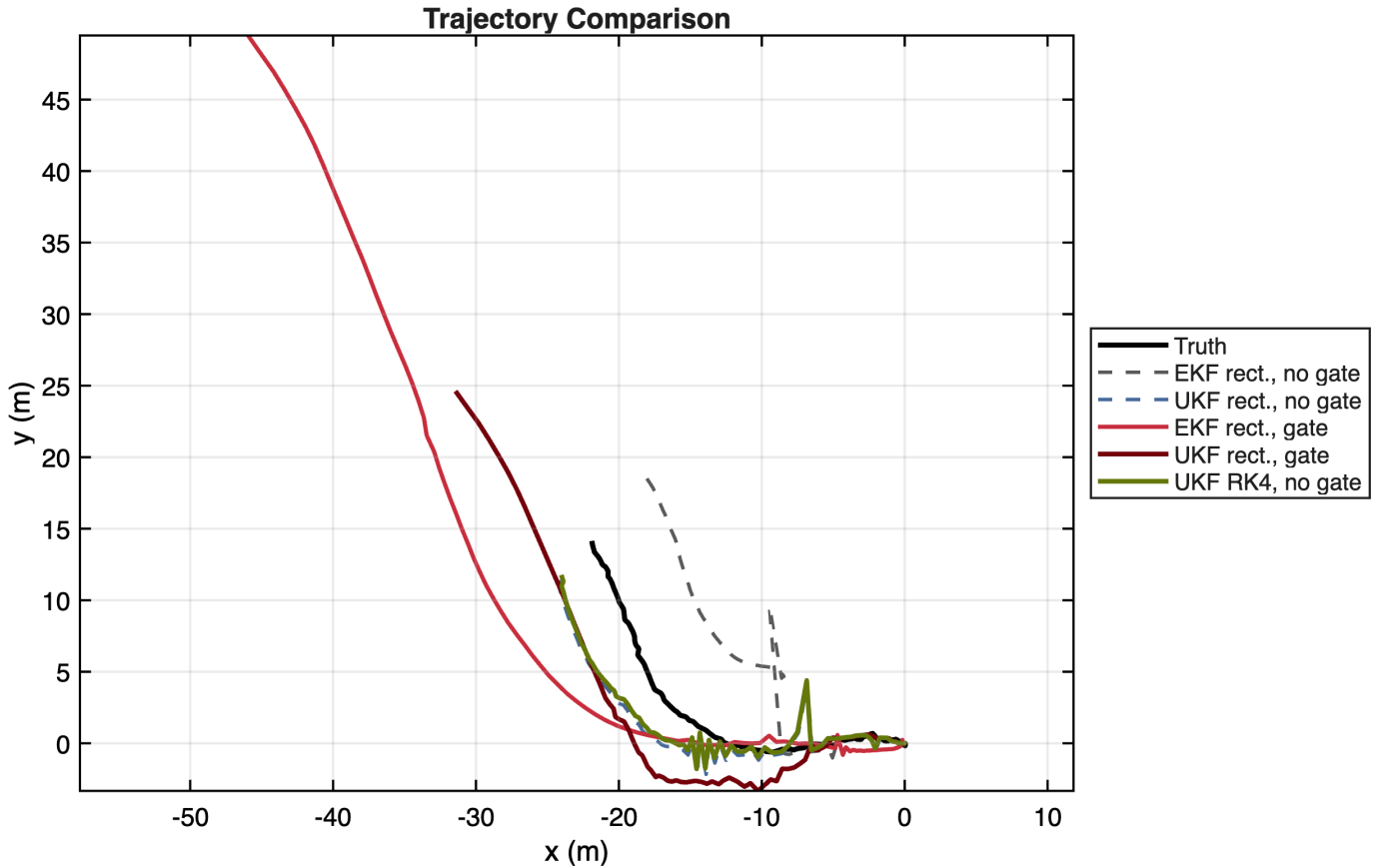


Fig. 2. Trajectory comparison for the rectangular EKF and UKF baselines, the gated rectangular filters, and the ungated RK4 UKF. The vanilla UKF and the ungated RK4 UKF remain closest to the truth trajectory, while the gated runs drift late in the record after repeated measurement rejection.

configuration. The EKF remained the cheaper estimator at 1.759 ms per run on average, while the rectangular UKF required 3.849 ms. That extra cost is expected because the UKF must propagate and recombine the sigma-point set at every step.

TABLE I

AVERAGE RUNTIME OVER SIXTY REPEATED EXECUTIONS FOR THE VANILLA RECTANGULAR EKF AND UKF.

Configuration	Mean ms	Std ms
EKF rect., no gate	1.759	0.182
UKF rect., no gate	3.849	0.081

IV. PROBLEM IV. MEASUREMENT GATING

Measurement gating was implemented as a scalar normalized innovation squared test on each sensor separately, exactly as requested in the handout. The study used a fixed 99% gate with threshold 6.6349. The gate did what it was supposed to do for the heading-like sensor. The EKF heading RMSE improved by 16.0203 deg and the UKF heading RMSE improved by 0.9017 deg after the strongest y_2 outliers were rejected. However, the position RMSE increased by 11.6843 m for the EKF and by 1.7155 m for the UKF because the gate also rejected many late y_1 measurements. The final rectangular gated runs rejected 73 and 19 range measurements, respec-

tively, which made both filters coast too long on the process model alone. Appendix A expands this into a threshold sweep and shows that the best heading gate and best position gate are not the same operating point.

V. PROBLEM V. UKF WITH RUNGE-KUTTA INTEGRATION

Replacing the rectangular UKF propagation with RK4 improved the ungated UKF accuracy. The rectangular UKF produced a position RMSE of 2.8483 m, while the ungated RK4 UKF reduced that value to 2.6457 m for an improvement of 0.2026 m. The heading RMSE remained nearly unchanged at 4.6895 deg for the rectangular UKF and 4.7116 deg for the RK4 UKF. The main benefit therefore appeared in the planar trajectory and state histories, where the higher-order integrator reduced accumulated discretization error.

VI. PROBLEM VI. RUNTIME OF THE RK4 UKF VERSUS THE RECTANGULAR UKF

The integrator runtime comparison again used sixty repeated executions after one warm-up pass per configuration. The rectangular UKF required 3.741 ms per run on average, while the RK4 UKF required 4.700 ms. The RK4 propagation therefore added 0.960 ms per run relative to the rectangular UKF. That extra cost is small compared with the position-accuracy gain reported in Problem V.

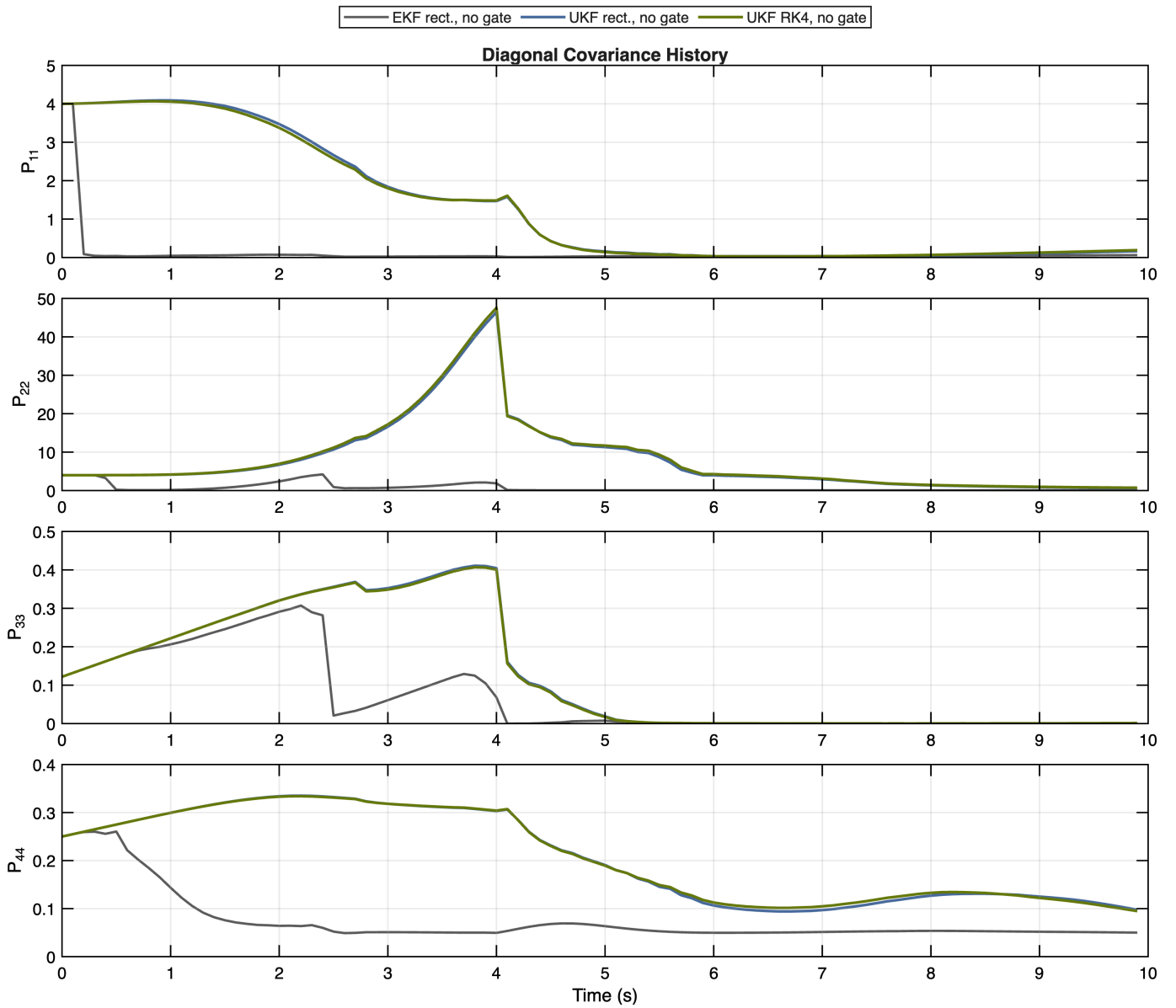


Fig. 3. Diagonal covariance histories for the rectangular EKF, rectangular UKF, and ungated RK4 UKF. The UKF variants retain visibly larger uncertainty than the EKF, which is consistent with their stronger nonlinear treatment.

TABLE II
AVERAGE RUNTIME OVER SIXTY REPEATED EXECUTIONS FOR THE RECTANGULAR
AND RK4 UKF IMPLEMENTATIONS.

Configuration	Mean ms	Std ms
UKF rect., no gate	3.741	0.023
UKF RK4, no gate	4.700	0.035

depends on the operating objective. A looser EKF gate can recover position accuracy, the two sensors are complementary rather than redundant, and UKF performance can improve further through prior and sigma-spread tuning.

Taken together, the six required comparisons point to a qualified conclusion. The vanilla UKF was the stronger baseline on this nonlinear problem, the scalar gate was useful for removing the worst heading-sensor outliers but harmed position accuracy when range updates were rejected too often, and RK4 was the cleanest algorithmic upgrade because it improved the ungated UKF trajectory with only a modest runtime increase. The appendix studies show that this conclusion

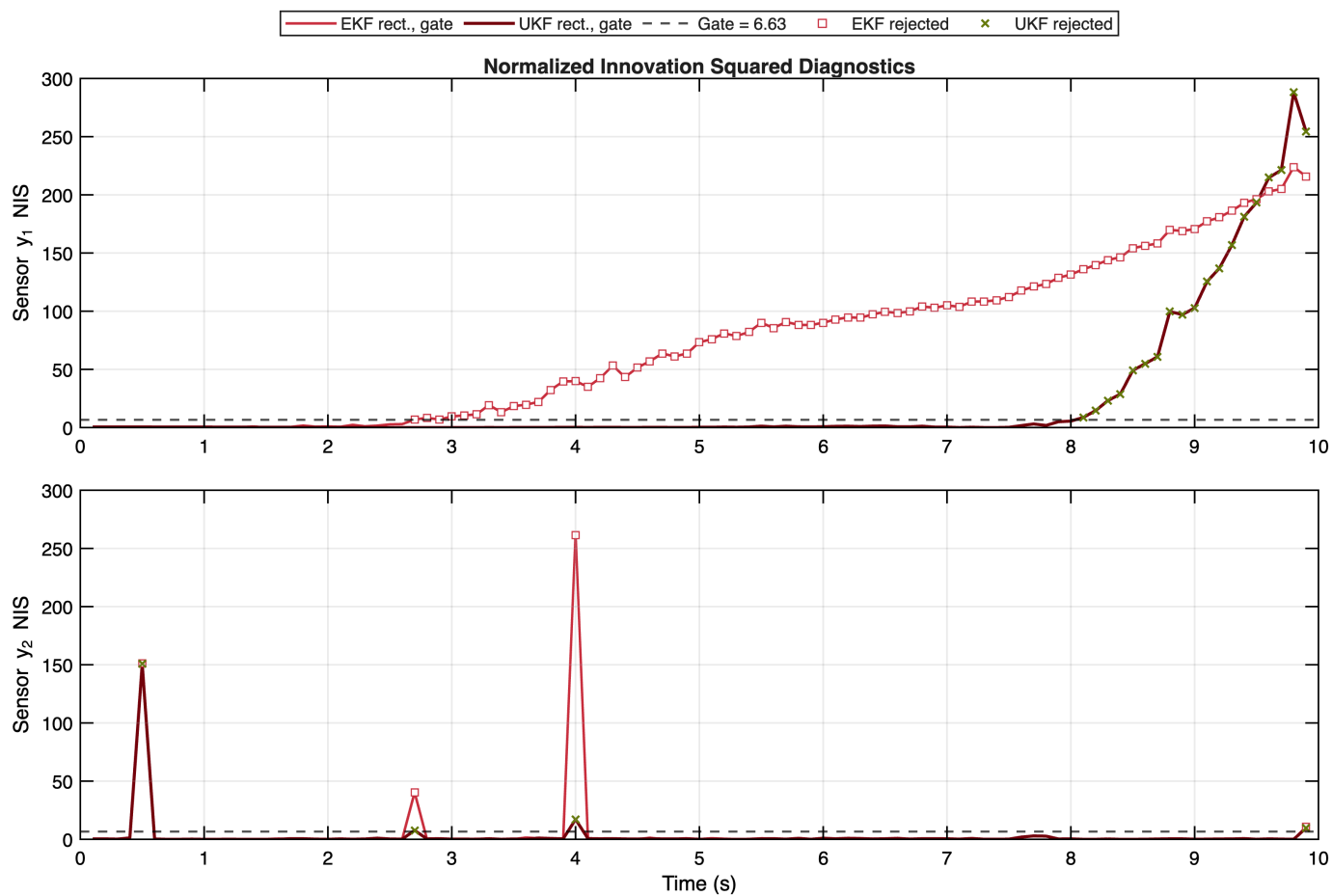


Fig. 4. Normalized innovation squared histories for both scalar sensors. The isolated y_2 spikes are clearly detectable, and the EKF and UKF rejection markers show how the late growth in y_1 residuals drives repeated range-measurement rejection.

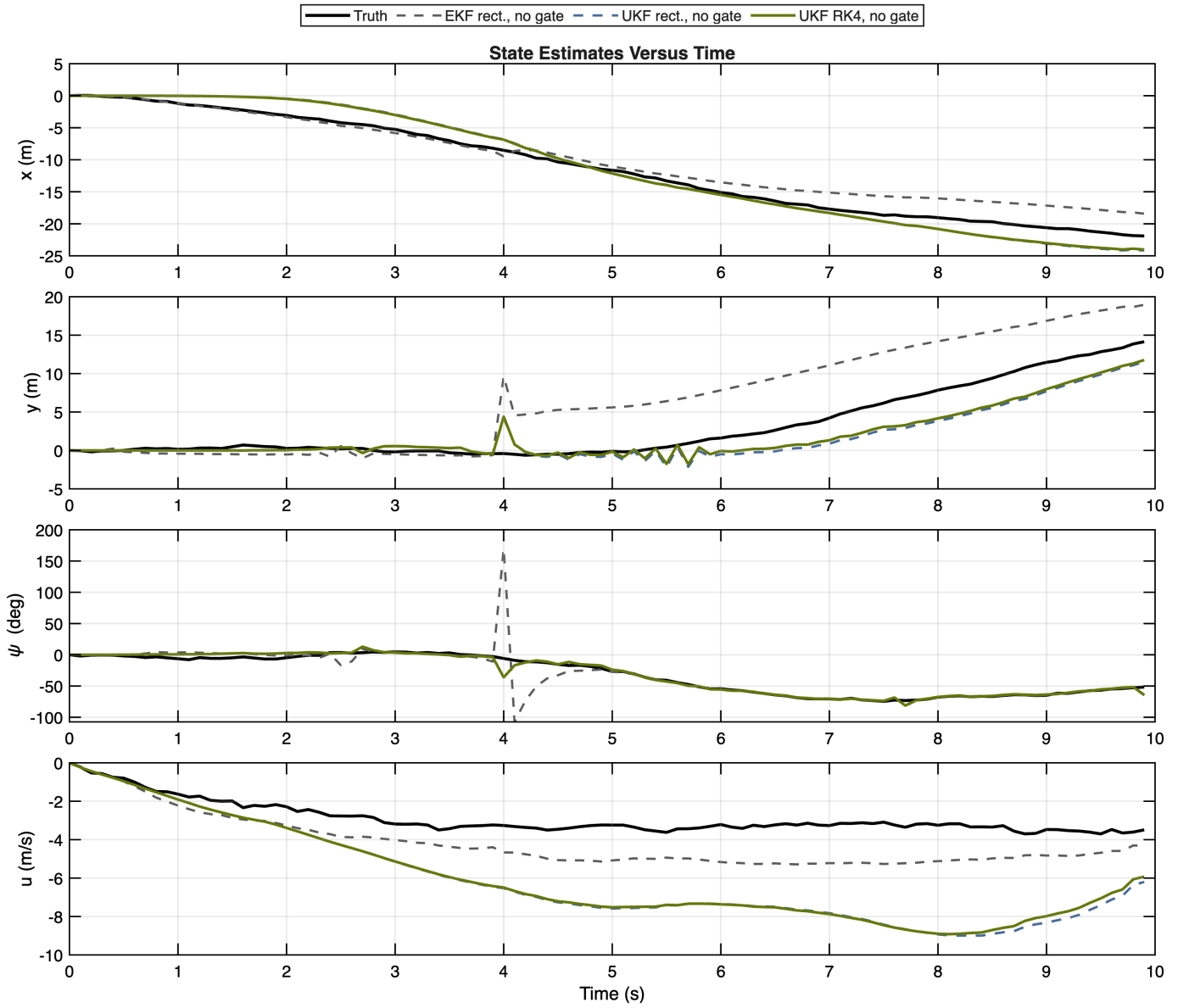


Fig. 5. State estimates versus time for the vanilla rectangular EKF, vanilla rectangular UKF, and ungated RK4 UKF. The RK4 UKF mainly improves the position channels while leaving the heading history close to the rectangular UKF result.

TABLE III
STATE-ESTIMATION ACCURACY, COVARIANCE SUMMARY, AND MEASUREMENT-REJECTION COUNTS FOR ALL EVALUATED CONFIGURATIONS.

Configuration	x	y	psi deg	u	pos.	avg tr(P)	rej y1	rej y2
EKF rect., no gate	1.9084	4.7656	22.1313	1.4067	5.1335	0.9432	0	0
UKF rect., no gate	1.7247	2.2667	4.6895	3.6135	2.8483	9.8452	0	0
EKF rect., gate	10.8599	12.8413	6.1110	6.4933	16.8178	3.5597	73	4
UKF rect., gate	2.8845	3.5367	3.7879	5.5241	4.5638	14.7626	19	4
UKF RK4, no gate	1.6755	2.0475	4.7116	3.5487	2.6457	10.0960	0	0
UKF RK4, gate	3.0439	3.5357	3.7544	5.5725	4.6654	15.1788	20	4

VII. APPENDIX A. GATE THRESHOLD SENSITIVITY

The main gating result uses a scalar 99% normalized-innovation-squared threshold, but the behavior is not fully described by that single operating point. This sweep varies the scalar confidence level while keeping the same rectangular EKF and UKF, the same calibration variances, and the same sequential one-sensor-at-a-time update order. The best position result in this sweep is EKF with the 99.9% gate at 2.5889 m, while the best heading result is UKF with the 99% gate at 3.7879 deg.

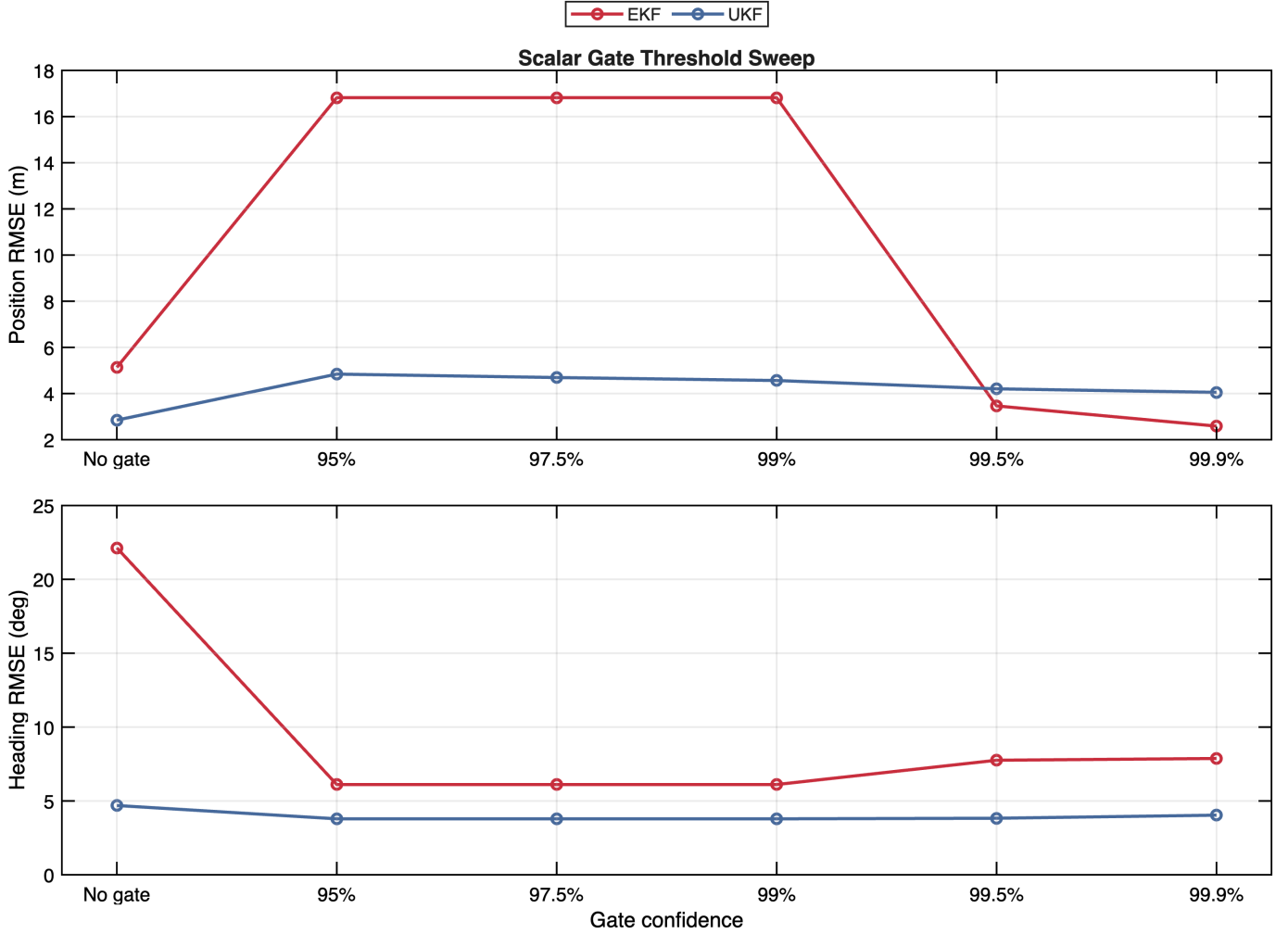


Fig. 6. Gate-threshold ablation for the rectangular EKF and UKF. Position accuracy and heading accuracy respond differently because rejecting late y_1 updates removes radial information, while rejecting isolated y_2 outliers can improve heading.

TABLE IV

GATE-THRESHOLD ABLATION METRICS FOR SCALAR NIS GATING. THE THRESHOLD COLUMN IS THE ONE-DEGREE-OF-FREEDOM CHI-SQUARE CUTOFF USED INDEPENDENTLY FOR EACH SENSOR.

Filter	Gate	Threshold	Pos. RMSE	Heading deg	u RMSE	avg tr(P)	rej y1/y2
EKF	No gate	inf	5.1335	22.1313	1.4067	0.9432	0/0
EKF	95%	3.8415	16.8178	6.1110	6.4933	3.5597	73/4
EKF	97.5%	5.0239	16.8178	6.1110	6.4933	3.5597	73/4
EKF	99%	6.6349	16.8178	6.1110	6.4933	3.5597	73/4
EKF	99.5%	7.8794	3.4633	7.7553	3.0603	1.2323	23/4
EKF	99.9%	10.8276	2.5889	7.8727	2.7683	1.2038	19/3
UKF	No gate	inf	2.8483	4.6895	3.6135	9.8452	0/0
UKF	95%	3.8415	4.8416	3.7884	5.5797	14.8075	21/4
UKF	97.5%	5.0239	4.6934	3.7882	5.5575	14.7811	20/4
UKF	99%	6.6349	4.5638	3.7879	5.5241	14.7626	19/4
UKF	99.5%	7.8794	4.2038	3.8281	5.3530	14.4930	17/3
UKF	99.9%	10.8276	4.0511	4.0355	5.2370	14.4700	15/2

The sweep makes the tradeoff in Problem IV explicit. A tighter gate can remove large innovations, but once the predicted trajectory drifts, the range innovation itself becomes large and the gate starts rejecting information that would otherwise pull the estimate back toward the measurement record. That feedback loop explains why the gated heading result can improve while the position result degrades.

VIII. APPENDIX B. SENSOR CONTRIBUTION ABLATION

The two sensors contribute different information. The range-like sensor y_1 directly constrains radial position but does not identify the sign or tangential component of the planar state by itself. The cubic heading sensor y_2 constrains orientation, but its local slope is $3\psi^2$, so it becomes weak near $\psi = 0$ and highly nonlinear as the heading grows in magnitude. This ablation reruns the rectangular EKF and UKF with both sensors, with only y_1 , with only y_2 , and with prediction only.

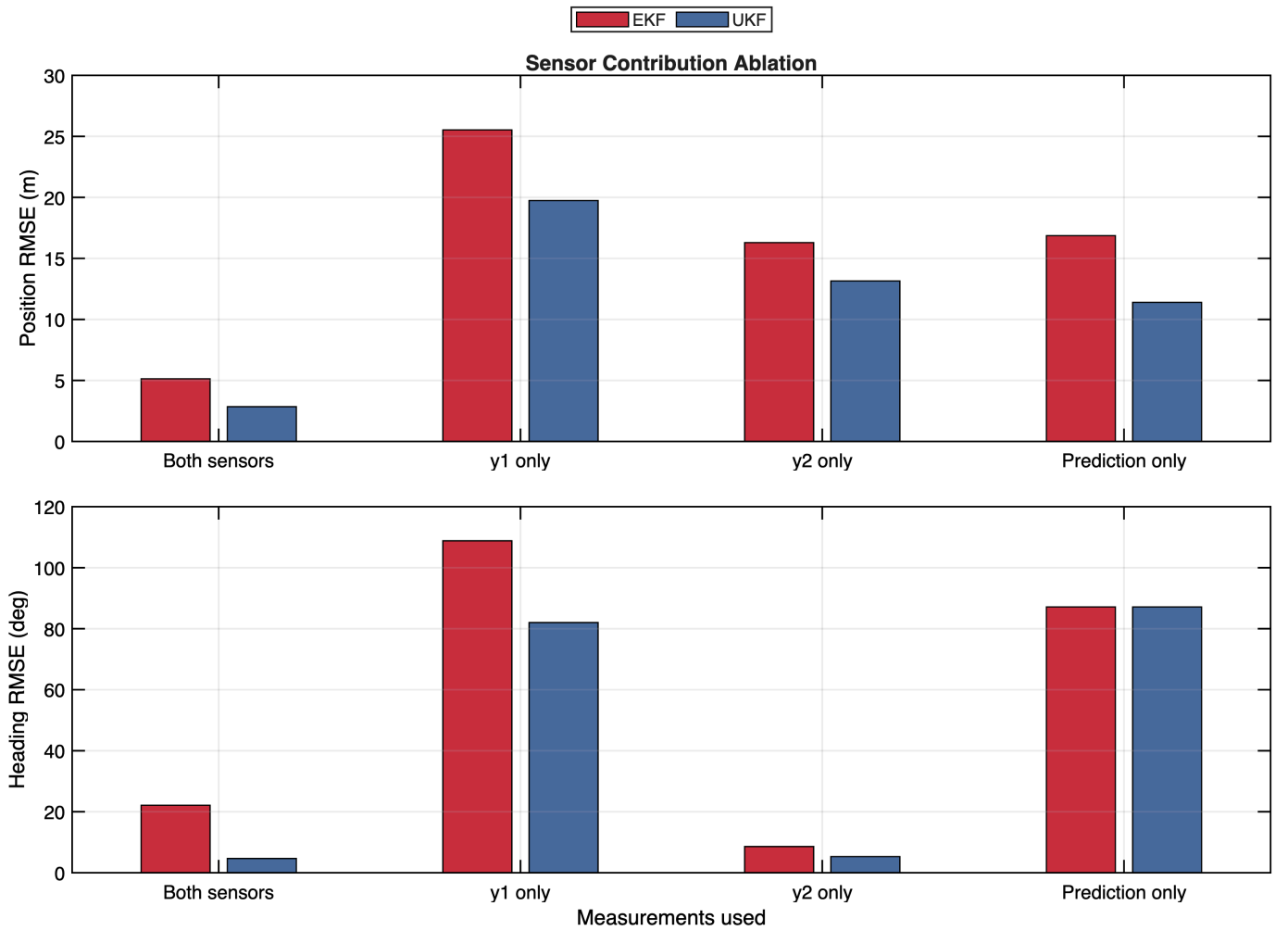


Fig. 7. Measurement-subset ablation for rectangular EKF and UKF runs. The comparison separates radial measurement information from heading measurement information and includes a prediction-only baseline.

TABLE V

SENSOR CONTRIBUTION ABLATION. THE BEST POSITION RESULT IS UKF USING BOTH SENSORS AT 2.8483 M, AND THE BEST HEADING RESULT IS UKF USING BOTH SENSORS AT 4.6895 DEG.

Filter	Measurements	Pos. RMSE	Heading deg	u RMSE	avg tr(P)	rej y1/y2
EKF	Both sensors	5.1335	22.1313	1.4067	0.9432	0/0
EKF	y1 only	25.5305	108.8226	6.1079	5.7131	0/0
EKF	y2 only	16.2937	8.6002	6.6772	17.8372	0/0
EKF	Prediction only	16.8624	87.1463	7.5095	186.3590	0/0
UKF	Both sensors	2.8483	4.6895	3.6135	9.8452	0/0
UKF	y1 only	19.7425	82.0128	6.5254	79.6766	0/0
UKF	y2 only	13.1532	5.3094	6.9164	43.3161	0/0
UKF	Prediction only	11.3987	87.1463	7.5095	214.3545	0/0

The useful baseline is not merely the result of one sensor dominating the other. The filters need y_1 to keep the path from drifting radially, and they need y_2 to keep heading errors from accumulating into position error. The UKF benefits more from the nonlinear measurement pair because it uses sigma-point propagation through $\sqrt{x^2 + y^2}$ and ψ^3 instead of a single local linearization.

IX. APPENDIX C. INITIAL COVARIANCE SCALE

The nominal runs use a zero-centered initial mean and an initial covariance with diagonal entries 4, 4, $(20 \frac{\pi}{180})^2$, and 0.25. This appendix scales that entire covariance by a single multiplier while keeping the initial mean, measurements, controls, and

rectangular integration fixed. The purpose is to test whether the EKF and UKF ranking depends on one hand-selected prior confidence.

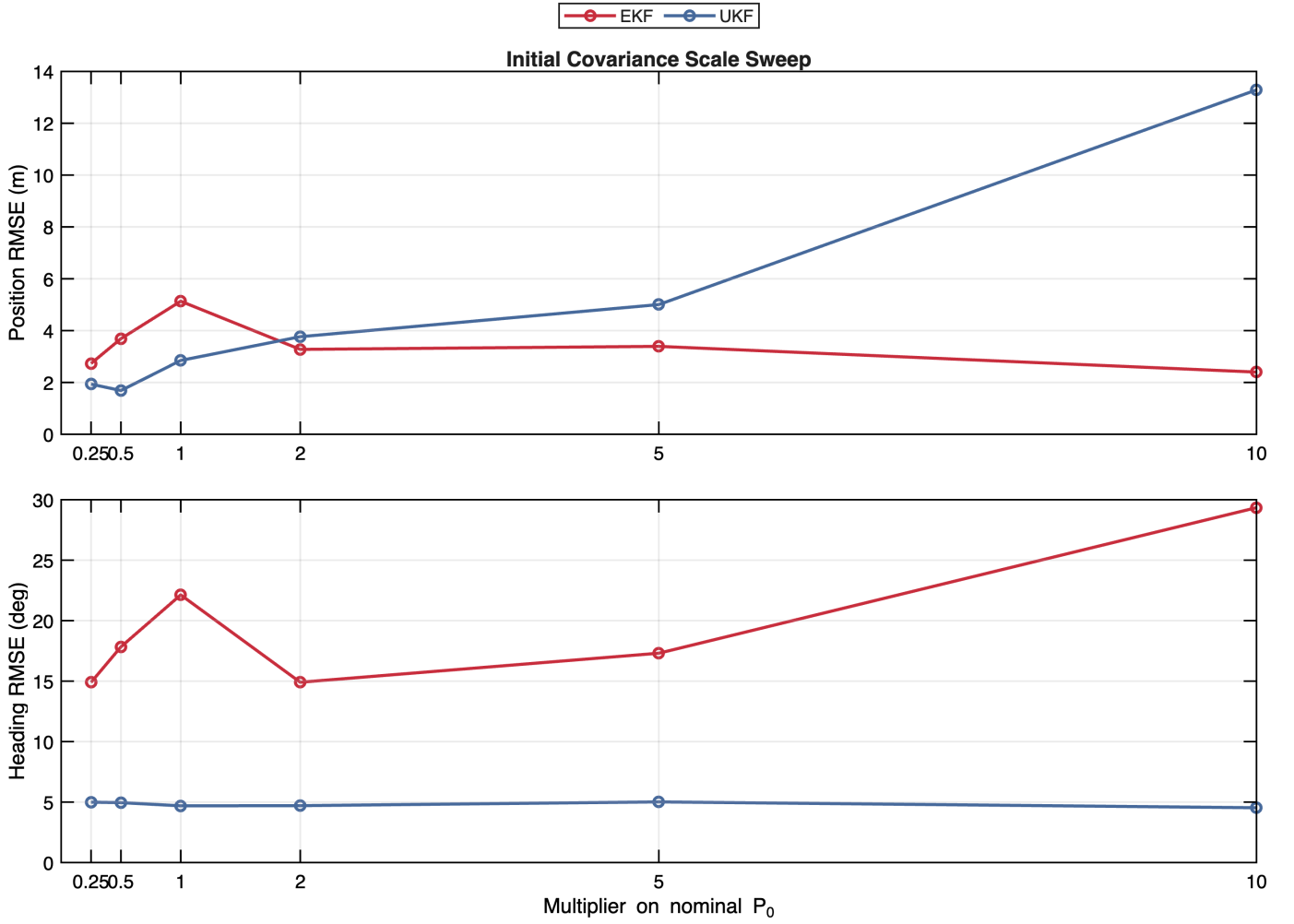


Fig. 8. Initial covariance scale ablation for the rectangular EKF and UKF. The horizontal axis multiplies the nominal P_0 used in the main study.

TABLE VI
INITIAL COVARIANCE SCALE ABLATION METRICS. THE BEST POSITION RESULT IS UKF AT A SCALE OF 0.5 WITH 1.6879 M POSITION RMSE.

Filter	P_0 scale	Pos. RMSE	Heading deg	u RMSE	avg tr(P)
EKF	0.25	2.7316	14.8990	1.6901	0.6158
EKF	0.5	3.6891	17.8381	1.5688	0.7417
EKF	1	5.1335	22.1313	1.4067	0.9432
EKF	2	3.2750	14.9076	1.5857	1.3990
EKF	5	3.3912	17.3085	1.6009	1.7238
EKF	10	2.3993	29.3424	1.8481	2.9515
UKF	0.25	1.9343	4.9912	2.8874	4.3863
UKF	0.5	1.6879	4.9532	2.9613	6.0709
UKF	1	2.8483	4.6895	3.6135	9.8452
UKF	2	3.7641	4.7030	4.0733	16.9142
UKF	5	5.0043	5.0142	3.7185	45.8353
UKF	10	13.2877	4.5313	3.9228	117.1644

The covariance scale sweep is a useful check because the chosen zero-centered prior starts at a point where the EKF measurement Jacobians for both sensors are locally weak or singular. A very small initial covariance can make that local

linearization too trusted, while a larger covariance gives the measurement updates room to move the estimate. The UKF is still not immune to prior tuning, but the sweep shows why maintaining uncertainty is preferable on this problem.

X. APPENDIX D. UKF SIGMA-POINT SPREAD

The main UKF uses $\alpha = 10^{-3}$, $\beta = 2$, and $\kappa = 0$. The parameter α controls the spread of sigma points around the mean, so it changes how aggressively the UKF samples the nonlinear measurement and process models. This appendix varies α for the rectangular UKF while leaving all other settings unchanged.

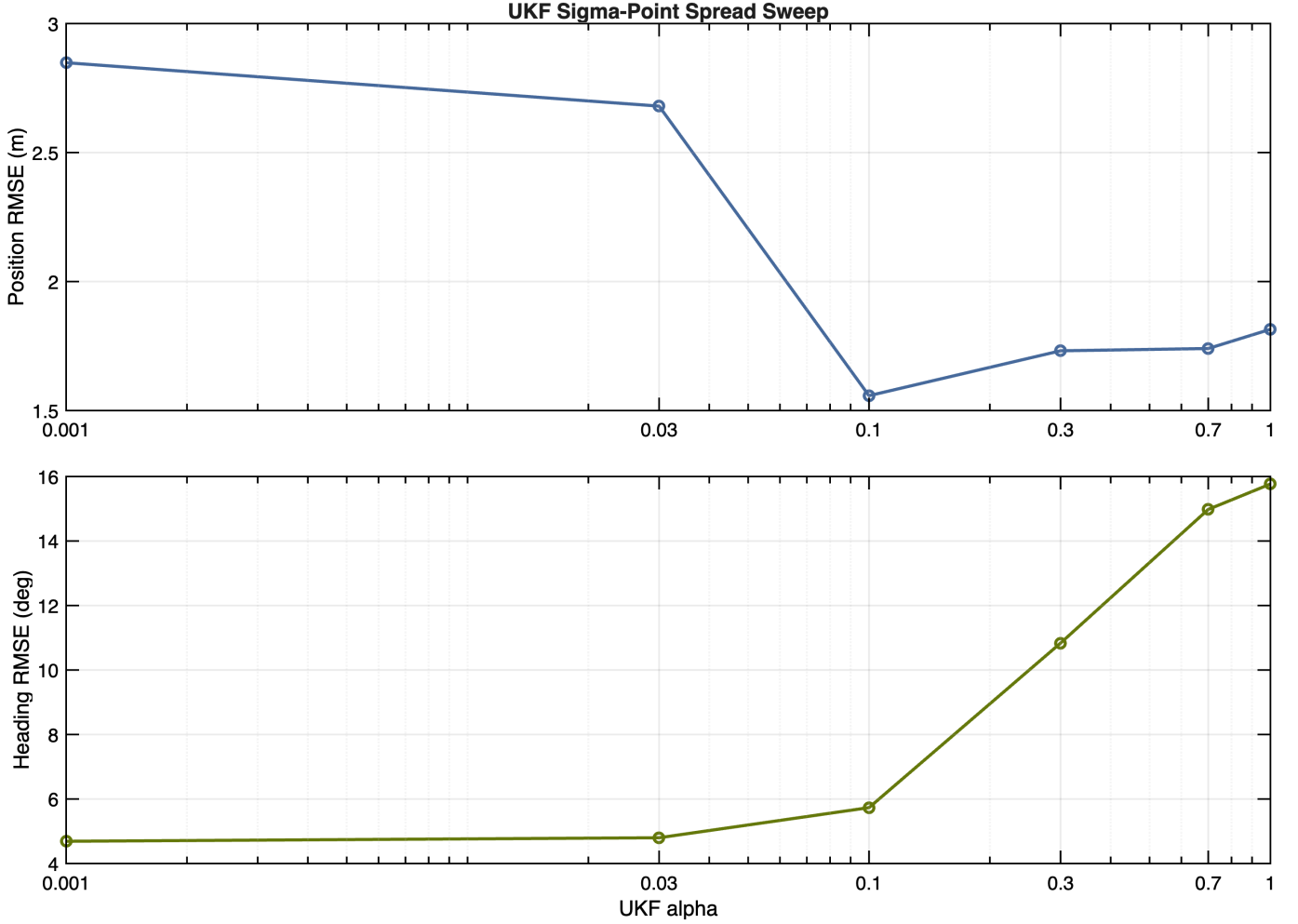


Fig. 9. UKF alpha ablation for the rectangular UKF. The sweep tests whether the reported UKF performance depends on the default narrow sigma-point spread.

TABLE VII

UKF SIGMA-POINT SPREAD ABLATION. THE BEST POSITION RESULT OCCURS AT $\alpha = 0.1$ WITH 1.5576 M POSITION RMSE, WHILE THE BEST HEADING RESULT OCCURS AT $\alpha = 0.001$ WITH 4.6895 DEG HEADING RMSE.

α	Pos. RMSE	Heading deg	u RMSE	avg tr(P)
0.001	2.8483	4.6895	3.6135	9.8452
0.03	2.6800	4.7987	3.5114	9.6139
0.1	1.5576	5.7322	2.6673	7.6227
0.3	1.7316	10.8339	2.7370	4.6149
0.7	1.7405	14.9804	2.8656	4.1723
1	1.8150	15.7675	2.9690	4.1409

The alpha sweep tests the numerical robustness of the UKF result. If the UKF advantage disappeared under modest alpha changes, the main conclusion would be more a tuning artifact than an estimator comparison. Instead, the sweep gives a tuning

envelope for the same algorithmic choice and clarifies how much performance is available from sigma-point spread alone before changing the integrator to RK4.